Optoelectronics- Solutions 2023

1. a) From the formula sheet, the integral forms of Faraday's law and Ampère's law are:
$\int_L \mathbf{E} \cdot d\underline{L} = -\iint_A \partial\mathbf{B}/\partial t \cdot d\underline{a}$
$\int_L \mathbf{H} \cdot d\underline{L} = \iint_A [\mathbf{J} + \partial\mathbf{D}/\partial t] \cdot d\underline{a}$
Each side of the equation contains an integral of different form, so the equations are normally transformed into differential form to make them easier to solve.

[1]

From the formula sheet, Stokes' Theorem states that:
$\int_L \mathbf{F} \cdot d\underline{L} = \iint_A \text{curl}(\mathbf{F}) \cdot d\underline{a}$
Applying Stokes' theorem to Faraday's Law:
$\int_L \mathbf{E} \cdot d\underline{L} = \iint_A \text{curl}(\mathbf{E}) \cdot d\underline{a} = -\iint_A \partial\mathbf{B}/\partial t \cdot d\underline{a}$
Since the area A is undefined, the two RH integrands must be equal, so:
$\text{curl}(\mathbf{E}) = -\partial\mathbf{B}/\partial t$

[2]

Applying Stokes' theorem to Ampère's Law:
$\int_L \mathbf{H} \cdot d\underline{L} = \iint_A \text{curl}(\mathbf{H}) \cdot d\underline{a} = \iint_A [\mathbf{J} + \partial\mathbf{D}/\partial t] \cdot d\underline{a}$
Since the area A is undefined, the two RH integrands must again be equal, so:
$\text{curl}(\mathbf{H}) = \mathbf{J} + \partial\mathbf{D}/\partial t$

[2]

b) For optical waves in dielectric media, we may assume:
$\mathbf{D} = \varepsilon_0\varepsilon_r\mathbf{E}$, since the medium is a dielectric
$\mathbf{B} = \mu_0\mathbf{H}$, since the medium is not magnetic
$\mathbf{J} = \sigma\mathbf{E} = 0$, since the conductivity is zero

[3]

Hence:
$\text{curl}(\mathbf{E}) = -\mu_0\partial\mathbf{H}/\partial t$
$\text{curl}(\mathbf{H}) = +\varepsilon_0\varepsilon_r\partial\mathbf{E}/\partial t$
Assuming now that fields vary at angular frequency ω, so $(\mathbf{E}, \mathbf{H}) = (\underline{E}, \underline{H}) \exp(j\omega t)$:
$\text{curl}(\underline{E}) = -j\omega\mu_0\underline{H}$
$\text{curl}(\underline{H}) = +j\omega\varepsilon_0\varepsilon_r\underline{E}$

[2]

c) Taking the curl of both sides of the upper equation:
$\text{curl}\{\text{curl}(\underline{E})\} = -j\omega\mu_0\,\text{curl}(\underline{H})$
Substituting using the lower equation to get:
$\text{curl}\{\text{curl}(\underline{E})\} = \omega^2\mu_0\varepsilon_0\varepsilon_r\underline{E}$

[2]

From the formula sheet:
$\text{curl}\{\text{curl}(\underline{F})\} = \text{grad}\{\text{div}(\underline{F})\} - \nabla^2\underline{F}$
If $\text{div}(\underline{E}) = 0$ (the other assumption) we then obtain the vector wave equation:
$\nabla^2\underline{E} + \omega^2\mu_0\varepsilon_0\varepsilon_r\underline{E} = 0$

[2]

d) The electric field of a particular plane electromagnetic wave can be written in time-dependent form as $\underline{E} = E_0 \exp(-jkz)\,\underline{j}$.

The divergence of a vector field $\underline{F}$ in cartesian coordinates is
$\nabla \cdot \underline{F} = \partial F_x/\partial x + \partial F_y/\partial y + \partial F_z/\partial z$
In this case, there is only a y-component, which is only a function of z, so:

$\nabla \cdot \underline{E} = 0$.

[2]

The curl of a vector field $\underline{F}$ in cartesian coordinates is:
$\nabla \times \underline{F} = (\partial F_z/\partial y - \partial F_y/\partial z)\,\underline{i} + (\partial F_x/\partial z - \partial F_z/\partial x)\,\underline{j} + (\partial F_y/\partial x - \partial F_x/\partial y)\,\underline{k}$
In this case, there is only a y-component, which is only a function of z, so:
$\nabla \times \underline{E} = (0 - \partial E_y/\partial z)\,\underline{i} + (0 - 0)\,\underline{j} + (0 - 0)\,\underline{k} = jkE_0\,\underline{i}$

[2]

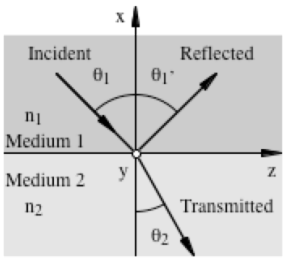
The results above imply that:
div($\underline{E}$) is indeed zero.

[1]

The magnetic field may be found from the expression for $\nabla \times \underline{E}$ and in this case only has an x-component. Since the electric and magnetic fields both have components perpendicular to each other and to the direction of propagation, the solution is a TEM wave.

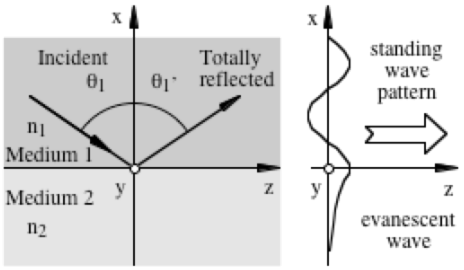
[1]

2. a) For incidence on a boundary between two dielectric media, there will in general be incident, reflected and transmitted waves as shown below.



[2]

When total internal reflection takes place there is no longer a propagating transmitted wave, and all the power is reflected. The field on the near side of the interface is a standing wave patter, and the field on the far side of the interface is an exponentially decaying wave.



[4]

b) Alhazen's law states that the angles θ_1 and θ_1' are equal.

[1]

Snell's law states that, the angles θ_1 and θ_2 for incidence at a boundary between two media of refractive indices n_1 and n_2 are related by $n_1\sin(\theta_1) = n_2 \sin(\theta_2)$

[1]

If the refractive index n_1 is greater than n_2, the transmitted wave angle θ_2 is greater than the incident wave angle θ_1. Total internal reflection occurs when θ_2 reaches $\pi/2$. This arises at a critical value of θ_1 when $n_1\sin(\theta_{1c}) = n_2$, so that:
$\theta_{1c} = \sin^{-1}(n_2/n_1)$

[2]

For a glass-air interface $\theta_{1c} = 41.83^o$, so $1/n_1 = \sin(41.83^o) = 0.666$ and $n_1 = 1.5$.

[1]

At normal incidence, the amplitude reflection coefficient is $\Gamma = (n_1 - n_2) / (n_1 + n_2)$
In this case, $\Gamma = (1.5 - 1) / (1.5 + 1) = 0.5/2.5 = 0.2$
The power reflection coefficient is then $0.2^2 = 0.04$, or 4%

[1]

c) A 2D parabolic index medium can be described by the refractive index variation:

$$n(x) = n_0 \sqrt{\{1 - (x/x_0)^2\}}$$

Here, n_0 and x_0 are constants, and x is the transverse distance from the axis of propagation.

Referring to the figure below: the local ray direction is given by $dx/dz = k_x/k_z$
Here $k_z = \beta$ is the modal propagation constant) and $|\underline{k}| = k_0 n$
Since $k_x^2 + k_z^2 = |\underline{k}|^2$ we get:
$k_x = \sqrt{(k_0^2 n^2 - \beta^2)}$, or $k_x = \sqrt{\{k_0^2 n_0^2 - \beta^2 - k_0^2 n_0^2 (x/x_0)^2\}}$

Hence, the equation for a ray trajectory is:
$\beta^2(dx/dz)^2 = k_0^2n_0^2 - \beta^2 - k_0^2n_0^2(x/x_0)^2$

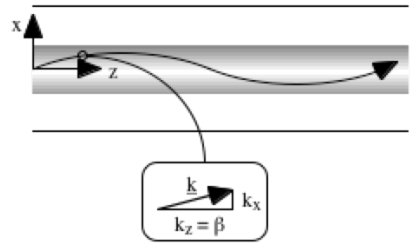
[3]

To prove that $x = A \sin(Bz + \psi)$ is a solution
Re-arrange the trajectory equation to get:
$k_0^2n_0^2 - \beta^2 - k_0^2n_0^2(x/x_0)^2 - \beta^2(dx/dz)^2 = 0$
Differentiate the solution to get:
$dx/dz = AB \cos(Bz + \psi)$
Substitute to get:
$k_0^2n_0^2 - \beta^2 - (k_0^2n_0^2A^2/x_0^2) \sin^2(Bz + \psi) - \beta^2A^2B^2 \cos^2(Bz + \psi) = 0$

A solution is possible if $\sin^2$ and $\cos^2$ terms add to unity.
This requires $k_0^2n_0^2A^2/x_0^2 = \beta^2A^2B^2$
The value of B can then be found as $B = k_0n_0/\beta x_0$

The trajectory equation then reduces to $k_0^2n_0^2 - \beta^2 - k_0^2n_0^2A^2/x_0^2 = 0$
The value of A can then be found as $A = x_0 \sqrt{(1 - \beta^2/k_0^2n_0^2)}$
Because constant values of A and B can be found, the solution is valid.

[3]

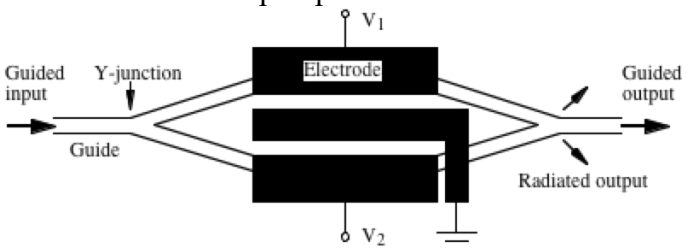


[1]

The solution above has the ray trajectory varying continually in a graded index medium. In contrast, zig-zag ray paths are obtained by abrupt reflections in step-index waveguides.

[1]

3. a) The figure below shows a Y-junction-based Mach-Zehnder interferometer, which consists of two back-to-back Y-junctions linked by straight waveguide sections containing one or more electro-optic phase shifters.

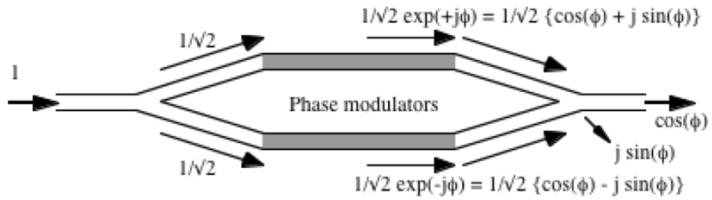


[4]

b) Operation of the device can be understood by assuming that:

- The LH Y-junction splits an input with unity amplitude into outputs with amplitude $1/\sqrt{2}$.
- The phase shifters (here, two) alter these to $1/\sqrt{2}\ \exp(\pm j\phi)$, where ϕ is the phase shift.
- These outputs can be expanded as $1/\sqrt{2}\ \{\cos(\phi) \pm j\ \sin(\phi)\}$
- The terms $\pm 1/\sqrt{2}\ \sin(\phi)$ represents an antisymmetric input to the RH Y-junction; consequently this component is radiated.
- The terms $+1/\sqrt{2}\ \cos(\phi)$ represents a symmetric input to the RH Y-junction; this component emerges with output amplitude $\cos(\phi)$
- The amplitude (and power) may therefore be controlled through the phase shift ϕ.

[4]



[2]

c) A solution may be found as follows.

- Assume the inputs are to ports 1 and 2 as shown below, with equal unity amplitudes.
- There can be no radiation loss at the first two Y-junctions, and their output amplitudes are all $1/\sqrt{2}$.

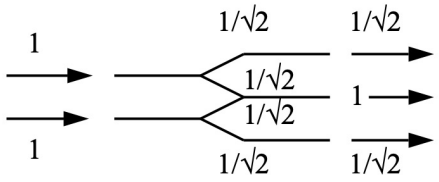
[3]

- The output amplitudes from the upper and lower guides (ports 3 and 5) are then $1/\sqrt{2}$.
- The third Y-junction sees equal inputs of $1/\sqrt{2}$. Since this is a symmetric input, there can be no radiation loss, and the inputs recombine as an output amplitude of 1 from port 4.

[3]

- Since light rays are reversible, input amplitudes of $(1/\sqrt{2}, 1, 1/\sqrt{2})$ at ports (3, 4, 5) will give output amplitudes of (1, 1) at ports (1,2).

[2]



d) Input amplitudes of $(1/\sqrt{2}, 1, 1/\sqrt{2})$ correspond to input powers of (1/2, 1, 1/2), corresponding to a total input power of 2. Output amplitudes of (1, 1) correspond to input powers of (1, 1). This also corresponds to a total output power of 2, so power is conserved.

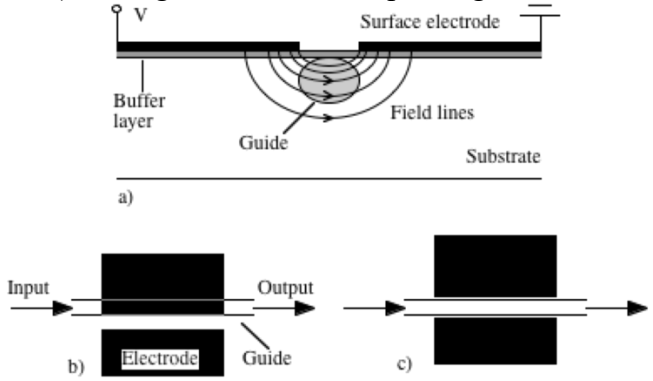
[2]

4. a) Titanium diffused lithium niobate phase modulators are preferred over carrier injection and electro-optic semiconductor modulators and direct modulation of a laser, since:

- Carrier injection modulators operate only up to a few hundred MHz
- Direct modulation of a semiconductor laser can only be achieved up to a few GHz
- Semiconductor and Ti:$LiNbO_3$ electro-optic devices can both operate up to at least 100 GHz, but
- Coupling losses to optical fibres are much lower for Ti:$LiNbO_3$ devices

[4]

b) Referring to the figure below: a) shows the cross-section of Ti:$LiNbO_3$ phase shifter; b) and c) show guide locations exploiting vertical and horizontal fields, respectively.



[3x2 = 6]

c) The change in effective index induced by a voltage V using an electro-optic phase modulator in a material of refractive index n with an electrode gap g is $\Delta n_{eff} = -n^3 r V \Gamma / 2g$. Here:

r is the electro-optic coefficient of the material and

[1]

Γ is the normalised overlap between the electric field of the guided mode and the electric field induced by the electrodes.

[2]

The phase shift obtained at wavelength λ in a device of length L is:
$\phi = 2\pi\Delta n_{eff}/\lambda$
To obtain a π radians phase shift, we then require:
$|2\pi\Delta n_{eff} L/\lambda| = \pi$
So that:
$|\Delta n_{eff} L| = n^3 r V L \Gamma / 2g = \lambda/2$
Hence:
$(VL)_\pi = \lambda g / n^3 r \Gamma$

[3]

d) Assuming that n = 2.2, r = 30.8 x 10^{-12} m/V, λ = 1.5 μm, g = 7 μm and Γ = 0.6:
$(VL)_\pi = 1.5 \times 7 / 2.2^3 \times 30.8 \times 0.6 = 0.0533$ Vm.

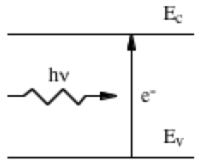
[2]

Hence, the drive voltages needed in devices of the lengths given are:
i) L = 10 mm: V = $0.0533/10^{-2}$ = 5.33 V
ii) L = 5 mm: V = 10.67 V

[2]

5. a) Photoconductive detectors operate by the absorption of photons in a semiconductor, which induces transitions from the valence to the conduction band if the photons have sufficient energy. The increase in carrier density thus obtained alters the resistance of the device, and hence a change output voltage in a potential divider circuit.

[2]



[2]

b) Assume incident beam of power P falls on detector.
If each photon has energy hf, their rate of arrival is $P/hf = P\lambda/hc$ (sec^{-1})
Assume each photon generates one electron transiting the detector.
The rate of generation of electrons is also $P\lambda/hc$ (sec^{-1})
Assume each electron carries a charge e
The ideal photocurrent is then $I_p = Pe\lambda/hc$ (A)
Assume fraction of photons that generate a detectable electron is η (quantum efficiency).
The actual photocurrent is then $I_p = P\eta e\lambda/hc$ (A)
The responsivity (current per unit optical power) is $R = I_p/P = \eta e\lambda/hc$ (mA/mW)

[4]

If $\eta = 0.6$ and $\lambda = 0.88$ μm:
$R = 0.6 \times 1.6 \times 10^{-19} \times 0.88 \times 10^{-6}/(6.62 \times 10^{-34} \times 3 \times 10^{8}) = 0.425$ mA/mW

[2]

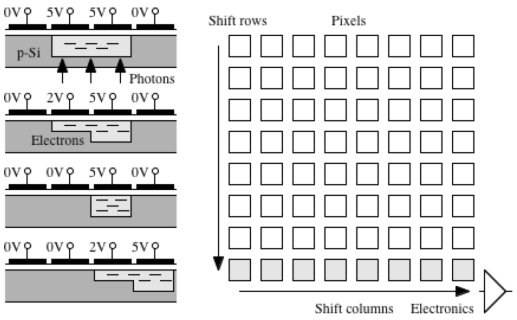
c) Photoconductive detectors have several drawbacks:

- Due to thermally generated carriers, the conductivity will not be zero when the beam is off**;** consequently, there will always be a DC voltage across R_L.
- The DC voltage may be blocked by a capacitor C**;** **h**owever, thermal carriers generate noise, and this cannot be blocked.
- Long lifetime is needed for high sensitivity. However, carriers then persist after beam is switched off.
- Consequently, difficult to combine high sensitivity with high speed.

[4]

d) A CCD array operates by photoconductive detection in channels created by the electric field of a surface electrode array. This field can have an additional component parallel to the surface, allowing photogenerated carriers to be passed along the channel for readout.

[2]



[2]

The drawbacks of photoconductive detectors are much less important in a CCD, because a) operating frequency is much lower, and b) low efficiency simply requires longer exposure.

[2]

6. a) The rate equations for a semiconductor laser can be written in the form:
$dn/dt = I/ev - n/\tau_e - G\phi(n - n_0)$
$d\phi/dt = \beta n/\tau_{rr} + G\phi(n - n_0) - \phi/\tau_p$

The physical processes described by the terms on the right-hand sides of these equations are:
$+I/ev$ Electron injection
$-n/\tau_e$ Loss of electrons by recombination
$-G\phi(n - n_0)$ Loss of electrons by stimulated emission, including absorption
$+\beta n/\tau_{rr}$ Creation of photons by spontaneous emission (β is coupling factor into stripe)
$+G\phi(n - n_0)$ Creation of photons by stimulated emission
$-\phi/\tau_p$ Loss of photons through end mirrors

[6]

There is no need for an additional rate equation for holes because electrons and holes are always created/destroyed in pairs.

[1]

b) In the steady state, the equations reduce to:
$I/ev - n/\tau_e - G\phi(n - n_0) = 0$
$\beta n/\tau_{rr} + G\phi(n - n_0) - \phi/\tau_p = 0$

Above threshold, spontaneous emission can be neglected, so the lower equation becomes:
$G\phi(n - n_0) - \phi/\tau_p = 0$
Substituting into the upper equation, $I/ev - n/\tau_e - \phi/\tau_p = 0$

Re-arranging, $\phi/\tau_p = I/ev - n/\tau_e$
Or $\phi/\tau_p = (I - I_{th})/ev$ where $I_{th} = nev/\tau_e$ is the threshold current and
Since $G\phi(n - n_0) - \phi/\tau_p = 0$ the electron density above lasing is $n = n_0 + 1/G\tau_p$

Now, the rate of emission of photons from the active volume v is:
$\Phi = v\phi/\tau_p = (I - I_{th})/e$
At wavelength λ, each photon carries energy hc/λ
Hence the optical power output is $P = hc\Phi/\lambda = hc(I - I_{th})/e\lambda$

[4]

The slope efficiency is $dP/dI = hc/e\lambda$
Assuming an energy gap of E (eV), $hc/\lambda = eE$ so $dP/dI = E$

[2]

c) The following data has been recovered from the student's logbook:

	I_1 (mA)	P_1 (mW)	I_2 (mA)	P_2 (mW)
Laser 1	12	1.0	20	9.0
Laser 2	18	2.8	25	12.8
Laser 3	18	8.5	24	17.0

If $P = (I - I_{th})\, dP/dI$ then $P_1 = (I_1 - I_{th})\, dP/dI$ and $P_2 = (I_2 - I_{th})\, dP/dI$
The slope efficiency can then be estimated from data pairs as $dP/dI = (P_2 - P_1) / (I_2 - I_1)$
Once dP/dI is known, the threshold current can be estimated as (say) $I_{th} = I_2 - P_2\, dI/dP$

[3]

The following values can then be estimated:

	I_1 (mA)	P_1 (mW)	I_2 (mA)	P_2 (mW)	dP/dI (mW/mA)	I_{th} (mA)
Laser 1	12	1.0	20	9.0	1	11
Laser 2	18	2.8	25	12.8	1.42	16
Laser 3	18	8.5	24	17.0	1.42	12

[3]

If lasers pass inspection only if their threshold current is below 13 mA and slope efficiency is above 1.3 mW/mA, shaded values indicate failure; only laser 3 is saleable.

[1]